

Kidney Stone Classification using Modified DenseNet for Feature Extraction and SVM for Classification

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Abstract—Kidney stones are a prevalent urological condition, and their accurate classification is essential for determining appropriate treatment strategies. This research introduces an innovative framework for kidney stone classification by integrating a modified DenseNet for feature extraction and classified with a Support Vector Machine (SVM).The proposed system leverages the DenseNet architecture, enhanced with modifications to optimize feature extraction specific to kidney stone imaging data. The model extracts detailed and hierarchical features from medical images, capturing fine-grained patterns associated with different types of kidney stones. These features are then input to SVM classifier, chosen for its robustness and effectiveness in handling high-dimensional data, to perform precise classification. Comparing to existing methods, this work is evaluated using benchmark kidney stone datasets, representing superior recitation in terms of classification accuracy, sensitivity, and specificity. A modified DenseNet and SVM integration ensures an efficient and scalable solution for kidney stone classification paving the way for improved diagnosis and personalized treatment planning in clinical applications.

Keywords—Kidneystone, DenseNet,SVM,Medical image processing,Classification,Deep learning

I.INTRODUCTION

The kidney or kidneys are the primary location for kidney stones. Urinary stones are the most prevalent urinary tract disorder and have affect humanity for generations, beginning around 4000 B.C. One major issue facing human health is the avoidance of kidney stone recurrence. A deeper comprehension of the mechanics underlying stone creation is necessary to prevent stone recurrence. Cardiovascular disease, chronic kidney disease, end-stage kidney failure, diabetes, and hypertension have all been linked together for kidney stones. Kidney stones have been proposed as a systemic condition associated with the metabolic syndrome. If that condition is related to nephrocalcinosis, it accounts for 2% to 3% of end-stage patients.

The symptoms of the kidney stone vary depending on the stone located. It may be located in kidney, ureter or bladder. No symptoms occur at first stage associated with stone production. The symptoms of stone illness are Haematuria (bloody urine), renal colic (severe cramping pain), flank pain (pain in the back), obstructive uropathy

(disease of the urinary system), urinary tract infections, obstruction of urine flow, and hydronephrosis (dilation of the kidney). In addition to the related discomfort from the stone incident, these diseases may cause vomiting and nausea [12]. As a result, medical care and lost work time come at a high cost and have an effect on both the economy and the standard of living.

Kidney stone classification is important because it helps doctors understand how and why a stone has formed, allowing them to choose the most suitable treatment approach. By organizing stones into clear categories, healthcare providers can plan effective management strategies, prevent complications, and reduce the chances of stones forming again. Classification also supports accurate diagnosis, helps monitor a patient's metabolic health, and guides long-term preventive care. Overall, it plays a vital role in improving patient outcomes and maintaining better kidney health.

II.LITERATURE SURVEY

Automated methods were used by Rajput et al. [14] to categorize kidney stones in CT scans. To detect automatically kidney stones without the need for human assistance, this research proposes an image processing method. The literature review and comparative analysis of the various algorithms accessible in the body of existing literature for the identification of urinary calculus in human bodies are also presented in this research.

S. M. B and colleagues [13] tackled the issue of kidney stones within the human body and suggested detection methodologies employing image processing techniques. Approaches such as morphological analysis, segmentation, and pre-processing were examined. The effectiveness of these techniques was assessed according to the resulting parameters and analyzed to determine the efficiency of the methods.

H. Dave and his team introduced an automatic ultrasonic object detection initiative in burn research domains, with current studies progressing in this area [12]. An application has been developed that enables doctors to identify the region of stones in CT images. The physician is required to select the area that the proposed system evaluates for stone presence. The extraction feature targets regions that might harbor stones. The overall accuracy of

the classification system is approximately 91%. The proposed system's complexity and accuracy will also be assessed through a confusion matrix.

L. Y. Myint et al developed an automatic unwanted object removing for kidney stone detection together with 3D visualization [11]. For the removal of surrounding unwanted objects, there are three steps in this proposed scheme. The initial phase involves eliminating hypodense and isodense areas through a process called intensity-based thresholding. In the following phase, size-based thresholding is applied to get rid of the bones in the abdominal region. This method achieves a sensitivity rate of 95.7%. To depict improved visual outcomes, the results from the simulation experiments are illustrated with 3D visualizations.

A. Soni and colleagues created an automated system for detecting stones using CT images. The Support Vector Machine, which is a type of learning model, is a potent method for classifying stones. It differentiates the vector spaces of kidneys affected by stones from those that are normal, dividing them into two distinct areas. When identifying the stones, image enhancements such as histogram adjustment and embossing may be referenced, this analytically assesses the variations in color.

A Back Propagation Network was developed by Akshaya et al. and used to identify kidney stones [9]. Decision making involves two key phases: 1. Feature extraction 2. Image classification. The process of feature extraction utilizes principal component analysis, while the classification of images employs a Back Propagation Network (BPN). This study introduces a segmentation method that utilizes the Fuzzy C-Mean (FCM) clustering technique.

Shahina and colleagues indicated that urine tests and X-ray examinations play a significant role in diagnosing kidney stones. The intensity and location of discomfort assist in making a clinical evaluation. Consequently, the diagnosis of kidney stone disease largely depends on the identification and assessment of kidney stones. In conjunction with feature extraction and morphological techniques, GAC segmentation is utilized to ascertain both the size and location of the kidney stones.

Shah and his team developed a model for renal calculi that included additional kidney-related areas and forecasted the existence of stones in kidney imaging. Computed Tomography captures X-ray images through necessary body slices. Initially, the images undergo multiple preprocessing steps, followed by segmentation, ultimately leading to the identification of the number of stones and their specific areas, which aids the physician in evaluating the patient's health condition.

The purpose of a research article authored by Thein et. al was create a pre-processing method for identifying and segmenting kidney stones in CT scans that does not rely on the reader's expertise [6]. To eliminate unwanted areas such as mats, soft tissues, and bones, a range of thresholding methods based on criteria like location, size, and intensity are implemented.

Thein et al intended to compare three distinct pre-processing methods aimed at eliminating noise from CT scans of kidney stones [5]. The findings reveal that the proposed techniques I, II, and III exhibit sensitivities of 90.91%, 92.93%, and 68.69%, correspondingly. The

average processing times for each method were recorded as 9.44 seconds, 10.14 seconds, and 34.5 seconds, respectively.

Akkasaligar et al employed the Level set segmentation approach to identify kidney stones [4]. Initially, the input images undergo pre-processing, followed by segmentation of the area of interest. The level set segmentation method proves effective in addressing segmentation challenges proficiently.

Hu et al conducted a study to demonstrate the potential of using X-ray dark-field tomography for distinguishing kidney stones [3]. A key benefit of this approach is its capability to visualize heterogeneous kidney stones, enabling the localization and identification of the various components within mixed-material kidney stones. A weighted total-variation regularization technique is employed to derive the dark-field to absorption signal ratio (DA Ratio) from projections that contain noise.

Vaish et al created a mobile application that allows for the automated diagnosis of kidney conditions based on CT imaging [2]. This application enables Smartphone to capture images from any CT scanner, process the acquired data, and deliver diagnostic outcomes. The automatic detection of kidney abnormalities utilizes the Viola Jones algorithm, incorporates texture feature extraction, and is further refined by an SVM classifier. The algorithm successfully identifies stones and cysts as the detected abnormalities. This application achieved an accuracy rate of 90.91% in recognizing these fault.

Ranjitha et al introduced RCD-AD (Renal Calculi Detection-Artifact Differentiation), in which fourteen distinct features are extracted and analyzed to provide insights into the issue that is crucial for extracting knowledge [1]. The GLCM (Gray Level Co-Occurrence Matrix) technique is utilized to extract features from the segmented Region of Interest (Renal Calculi/Artifact), acting as a statistical approach to extract second-order textural features.

III. PROPOSED SYSTEM

The proposed system integrates a modified DenseNet for extracting features along with a Support Vector Machine (SVM) for categorizing kidney stones. Modifications to the DenseNet architecture specifically designed for kidney stone imaging ensure the best possible extraction of hierarchical and fine-grained information. By identifying intricate patterns associated with different types of kidney stones, the extracted features bolster the model's effectiveness in differentiating among them. The features, once obtained, are fed into the SVM classifier, which was selected for its proficiency in managing high-dimensional information and delivering precise classifications. To ensure the system's efficacy and resilience, benchmark kidney stone datasets are used for both training and evaluation. The suggested method improves classification accuracy, sensitivity, and specificity by utilizing the combined advantages of DenseNet and SVM, making it a reliable and scalable solution for kidney stone classification. This system contributes to improved diagnostic precision and supports personalized treatment strategies in clinical settings.

1. Data Preprocessing

The first module involves preprocessing kidney stone imaging data to enhance its quality for feature extraction. This comprises noise reduction, contrast enhancement, and normalization techniques for standardizing input images. Techniques for augmenting data, including rotation, flipping and scaling are applied to enhance the ability of the model to generalize and minimize overfitting.

2. Feature Extraction using Modified DenseNet

This module extracts deep, hierarchical features from kidney stone images using a modified DenseNet architecture. To improve feature representation, the DenseNet model is optimized with particular changes, like modified convolutional layers or attention algorithms. The fine-grained patterns that differentiate various kidney stone types are captured by the retrieved characteristics.

3. Dimensionality Reduction and Feature Selection

Feature selection strategies are used to eliminate duplicate and irrelevant features in order to maximize classification performance. Dimensionality reduction methods like Principal Component Analysis (PCA) are utilized to lower computational demands and retain the most significant features.

4. Classification using Support Vector Machine (SVM)

The developed characteristics are subsequently directed to the SVM classifier, which was chosen for its capability in dealing with high-dimensional data. The SVM model classifies kidney stones by analyzing the characteristics that have been extracted, resulting in elevated accuracy, sensitivity, and specificity. To enhance classification performance, an appropriate kernel function, such as RBF or polynomial, is chosen.

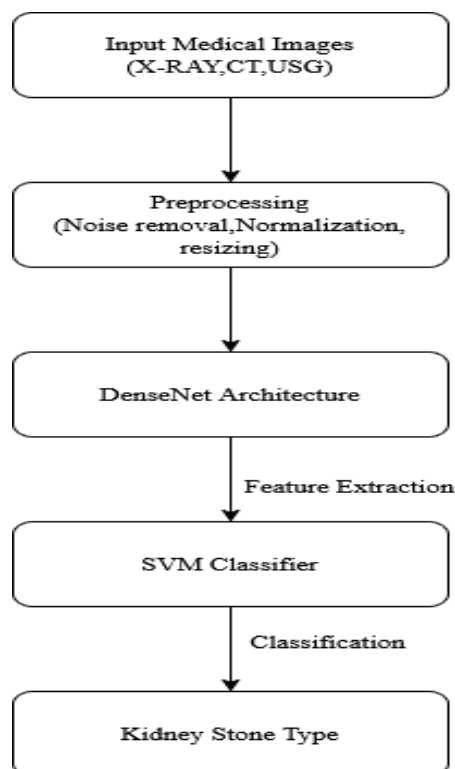


Fig 1: Modules of the Proposed System

5. Performance Evaluation

The concluding module focuses on assessing the suggested system by utilizing established kidney stone datasets. A range of performance indicators, such as accuracy in classification, sensitivity, specificity, and the F1 score, are calculated to evaluate the model's efficiency. A comparative evaluation with current techniques is performed to illustrate the advantages of the suggested method.

The fig 1. describe the stages of proposed modules. The initial phase of the proposed system involves the preprocessing of images related to kidney stones, applying techniques aimed at reducing noise, enhancing contrast, and normalizing images to boost overall quality. Additionally, data augmentation strategies, including rotation, flipping, and scaling, are applied to strengthen the model's generalization and durability. After preprocessing, a revised DenseNet structure is implemented within the feature extraction component. Better feature representation for classification is ensured by the changes made to DenseNet, which improve its capacity to capture hierarchical and fine-grained features unique to kidney stone patterns.

To keep the most relevant characteristics while reduced computational complexity, the system uses feature selection and dimensionality reduction techniques after feature extraction. By ensuring that only the most important patterns are classified, this phase enhances the system's efficiency. A Support Vector Machine (SVM) classifier receives the improved features after which it is able to handle high-dimensional data and correctly categorize kidney stones into various categories. Classification performance is further optimized by selecting a suitable kernel function, such as the Radial Basis Function (RBF).

Ultimately, the system is tested using standard kidney stone datasets, and its performance is evaluated based on classification accuracy, sensitivity, specificity, and F1-score. The strengths of the proposed technique are emphasized by comparing it against existing methods. The integration of a modified DenseNet for feature extraction alongside SVM for classification enables improved diagnosis and tailored treatment plans, ensuring a dependable, precise, and scalable solution for kidney stone classification.

IV. RESULT & DISCUSSION

The fig 2. demonstrates the input image of stone identification that is sourced from the Kaggle website. In this proposed work, we assembled a collection of 200 kidney images from the Kaggle platform, that were divided into two categories: normal and stone. The images were split into a training set consisting of 80 images and a testing set containing of 20 images.

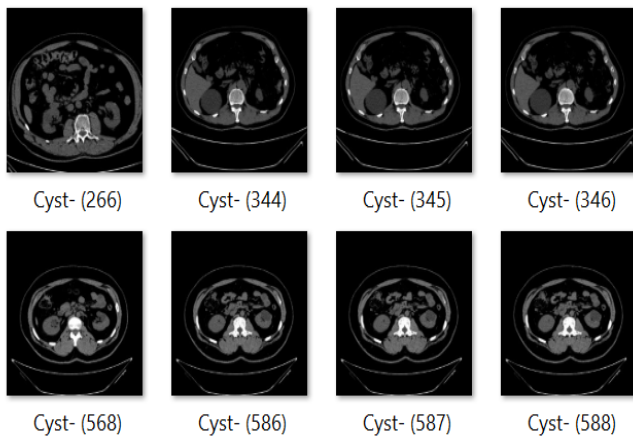


Fig 2: Input image

Next, we applied the Modified DenseNet architecture for training on the training set using a transfer learning methodology, whereby the pre-existing weights of the Modified DenseNet model served as the starting point for our training. We refined the Modified DenseNet model on the training set for 50 epochs, utilizing a batch size of 10 and a learning rate set at 0.0001.

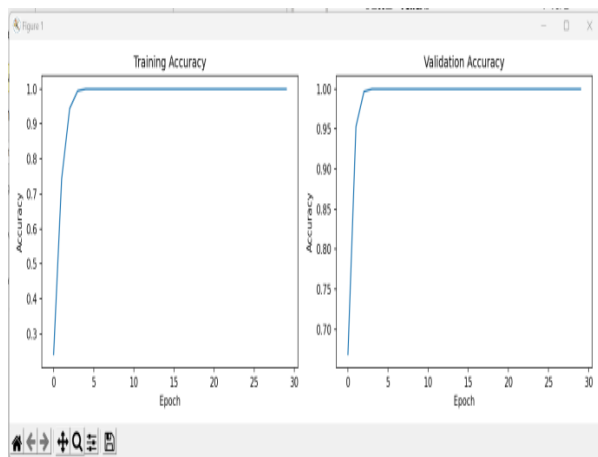


Fig 3: Validation and Testing curve

The fig 3. shows during the training process, the proposed model demonstrated its ability to efficiently learn from the provided dataset by steadily enhancing both training and validation accuracy. The ability of the model reduce prediction errors and maximize performance throughout multiple epochs was demonstrated by the consistent decline in training and validation loss observed simultaneously.

Notably, the validation accuracy and loss curves displayed remarkable stability and convergence, reflecting the model's robust generalization to unseen data. This is extremely important in medical image analysis, where clinical decision-making depends on the model's ability to correctly classify cases of cardiomegaly in real-world situations. Our suggested deep learning method provides a non-invasive and effective way to detect stones in CT images, and the patterns in the curves show how reliable and effective it is.

The advantages of our suggested model are further supported by comparison with other models. Our model performed better in classifying cardiomegaly than models

such as GoogleNet, InceptionNet, and ResNet, as demonstrated by its significantly higher accuracy and lower loss. This implies that our model's effectiveness in medical image analysis tasks is greatly influenced by the architectural decisions, optimization methods, and data preprocessing strategies used.

The observed results hold substantial implications for clinical practice, as accurate and efficient cardiomegaly detection can lead to timely intervention and improved patient outcomes. By leveraging deep learning and convolutional neural networks, our research aids in advancing automated tools for identifying cardiac irregularities, facilitating early diagnosis and intervention in cardiovascular medicine.

Table 1: Comparison chart

Parameter	F-score	Precision	Recall	Accuracy
GoogleNet	0.8	0.72	0.83	0.84
InceptionNet	0.81	0.86	0.85	0.86
ResNet	0.9	0.84	0.88	0.88
U-Net	0.9	0.89	0.87	0.90
M-Unet	0.82	0.75	0.87	0.93
Proposed	0.91	0.87	0.89	0.94

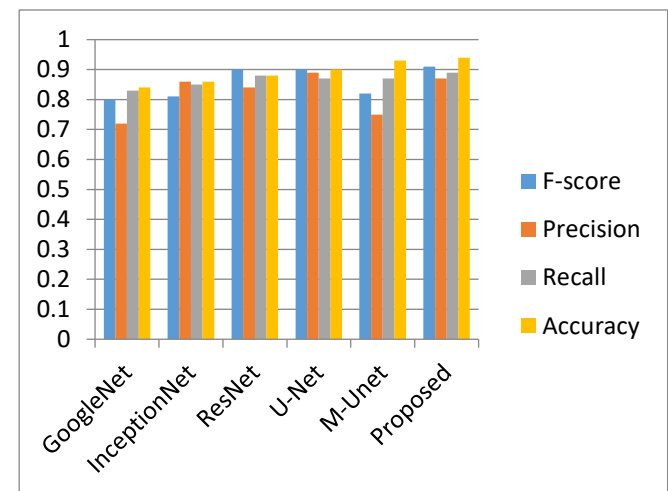


Fig 4: Performance chart

Overall the table 1 shows the highest F-score, precision, and recall among the six models, indicating better performance regarding trade-off between precision and recall, as well as positive results. The proposed model performs slightly lower accuracy than GoogleNet, InceptionNet, ResNet and U-Net but higher than M-Unet. The recital of a model depend on different factors like the dataset, model architecture, hyper parameters, and evaluation metrics used, and further analysis may be required to draw definitive conclusions about the model's performance. Fig 4 performance chart for proposed method with older bench mark datasets.

V.CONCLUSION

The proposed system an effective and robust framework for kidney stone classification was developed by integrating a modified DenseNet architecture for feature extraction with an SVM classifier for final prediction. The enhanced DenseNet successfully captured fine-grained textures and hierarchical representations inherent in kidney stone CT images, while the SVM demonstrated strong discriminative capability in handling the resulting high-dimensional feature space. Experimental results obtained from benchmark datasets confirmed that the proposed method outperforms existing approaches reported in the literature, particularly in terms of accuracy, precision, recall, and F-score. Comparative analysis with state-of-the-art models such as GoogleNet, InceptionNet, ResNet, U-Net, and M-UNet further validates the improved performance and generalization capability of the proposed system. Overall, the findings indicate that combining deep feature extraction with classical machine learning provides a reliable and computationally efficient solution for automated kidney stone classification. This framework presents a scalable and reliable result for kidney stone diagnosis, contributing to better clinical decision-making and customized treatment planning. Future work may explore expanding the dataset, integrating multi-modal imaging, and deploying the model in real-time diagnostic environments to further enhance clinical applicability.

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